

Homework 3

MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. Given the following twist ξ , what is the homogeneous transformation matrix?

(a) $\xi = [\mathbf{v}^\top \ \boldsymbol{\Omega}^\top]^\top = [3/\pi \ 0 \ 0 \ \pi/3 \ 0 \ 0]^\top$.

(b) $\xi = [\mathbf{v}^\top \ \boldsymbol{\Omega}^\top]^\top = [0 \ 0.5 \ 0 \ 0 \ \pi/2 \ 0]^\top$.

(c) $\xi = [\mathbf{v}^\top \ \boldsymbol{\Omega}^\top]^\top = [1 \ 0 \ 0 \ 0 \ 0 \ \pi/3]^\top$.

Problem 2. Given the homogeneous transformation matrix,

$$\mathbf{T} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) What is the displacement?
- (b) What is the Quaternion that encodes the rotation?
- (c) What is the Angle-Axis that encodes the rotation?
- (d) What is the twist ξ that encodes the full transform?

Problem 3. For the following give one pro and one con for using it as a representation,

- (a) Rotation and displacement separately.
- (b) Homogeneous transformation matrices.
- (c) Twist coordinates.

Problem 4. Given the following two homogeneous transformations, use MATLAB to plot a linear interpolation between the two. Your plot should contain a minimum of 5 intermediate points and should have the orthogonal axis (columns of the rotation matrix) drawn at each point. *Hint:* Use the `quiver3` function.

$$\mathbf{T}_{\text{initial}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{T}_{\text{final}} = \begin{bmatrix} 0.5000 & -0.6124 & 0.6124 & 0.8415 \\ 0.6124 & 0.7500 & 0.2500 & 0.3251 \\ -0.6124 & 0.2500 & 0.7500 & -0.3251 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- (a) Perform the interpolation using ZYZ angles and position.
- (b) Perform the interpolation using Angle-Axis and position.
- (c) Perform the interpolation using Quaternion and position (remember to enforce the length constraint on the quaternion before calculating the intermediate orientations).
- (d) Perform the interpolation using twist.
- (e) Comment on the similarities or differences.